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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech. End Semester Theory Examinations, November/December 2023.

Fifth Semester

Computer Science and Engineering

UCS2503 SOFTWARE ENGINEERING

(Regulations 2021)

Time: **Three Hours**Maximum: **100 Marks**

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

CO1:	Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
CO2:	Identify and analyze the requirements and construct their models (K5)
CO3:	Apply systematic procedure for software design (K3)
CO4:	Compare and contrast the various testing and maintenance activities (K2)
CO5:	Make use of agile development and Devops (K3)
CO6:	Identify unethical issues and apply ethical practices for a given case study (K3)

Part – A (5 × 2 = 10 Marks)

		KL	CO	PI
1.	How the productivity and cost are related to project estimation? Explain.	K2	CO1	1.3.1 2.1.2
2.	What is the purpose of a feasibility study?	K1	CO2	1.4.1 2.1.2
3.	How fault tolerance is designed in decentralized architecture?	K1	CO3	3.4.1 13.3.2
4.	List some coding practices followed in software companies.	K1	CO4	3.4.1 13.3.2
5.	What do you mean by fixing bugs in agile development?	K1	CO5	1.3.1

Part – B (5 × 6 = 30 Marks)

			KL	CO	PI	
6.	With the project data given below draw the network diagram and calculate the total time taken to complete the project.		K3	CO1	1.3.1 1.4.1	
	Task	Predecessor				Duration(in days)
	A	-				2
	B	A				2
	C	B				1
	D	A				1
	E	B,D				4
	F	C,E				2

7.	Develop the use case diagram for the following case study. A Coffee Vending Machine dispenses coffee to customers. Customers order coffee by selecting a recipe from a set of recipes. Customers pay for the coffee using coins and the change is given back, if any, to the customers. The Service Assistant loads ingredients (coffee-powder, milk, sugar, water and chocolate) into the coffee machine. The service assistant selects a recipe by indicating the name of the coffee, the units of coffee powder, milk, sugar, water and chocolate to be added as well as the cost of the coffee.	K3	CO2	3.1.2		
8.	Apply the refactoring to the following code and explain the importance of refactoring. <table><tr><td><u>Before refactoring:</u> while True: mass = int(input("Enter the mass value: ")) if mass > 0: break while True: acceleration = int(input("Enter the acceleration: ")) if acceleration > 0: break print("The Force is", mass * acceleration)</td><td><u>After refactoring :</u> ?</td></tr></table>	<u>Before refactoring:</u> while True: mass = int(input("Enter the mass value: ")) if mass > 0: break while True: acceleration = int(input("Enter the acceleration: ")) if acceleration > 0: break print("The Force is", mass * acceleration)	<u>After refactoring :</u> ?	K3	CO3	2.2.1 13.3.2 14.1.2
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9.	Construct flow graph and do the basis path testing for the simple subtraction pseudo code given below: (a) Program 'Simple Subtraction' (b) Input(x; y) (c) Output(x) (d) Output(y) (e) If x > y then DO (f) z = x - y (g) Else z = y - x (h) EndIf (i) Output(z) (j) Output "End Program"	K3	CO4	2.2.1 13.3.2 14.1.2		
10.	Explain branches and pull requests of the scrum model.	K2	CO5	1.3.1		

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI																											
11.	Assume that you are a software engineering consultant and have been called in by the vice-president for finance of a corporation that manufactures tires and sells them via its large chain of retail outlets. She wants your organization to build a product that will monitor the company's stock, starting with the purchasing of the raw materials and keeping track of the tires as they are manufactured, distributed to the individual stores, and sold to customers. a. Identify a suitable life cycle model for this product development and write your reason in selecting a life-cycle model for the project? (4) b. List any four risks involved in developing the software and how you attempt to mitigate each risk? (8)	K3	CO1	1.3.1 2.1.2																											
(Or)																															
12.	A target product has 8 simple inputs, 3 average inputs, and 11 complex inputs. There are 57 average outputs, 9 simple inquiries, 13 average master files, and 18 complex interfaces. a. Explain function point estimation procedure and apply it to the above product to estimate its size. Note: Use Table 1 weighting factor for estimation. (8) b. Differentiate LOC and function point estimation and why do you think that, despite its drawbacks, lines of code (LOC or KDSI) is so widely used as a metric of product size? (4) <table><tr><th rowspan="2">Component</th><th colspan="3">Level of Complexity</th></tr><tr><th>Simple</th><th>Average</th><th>Complex</th></tr><tr><td>External Inputs</td><td>3</td><td>4</td><td>6</td></tr><tr><td>External outputs</td><td>4</td><td>5</td><td>7</td></tr><tr><td>External Inquiries</td><td>3</td><td>4</td><td>6</td></tr><tr><td>Internal logical files</td><td>7</td><td>10</td><td>15</td></tr><tr><td>External interface files</td><td>5</td><td>7</td><td>10</td></tr></table> Table 1: Weighing factor	Component	Level of Complexity			Simple	Average	Complex	External Inputs	3	4	6	External outputs	4	5	7	External Inquiries	3	4	6	Internal logical files	7	10	15	External interface files	5	7	10	K3	CO1	1.3.1 2.1.2
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13.	Case study: When the customer wants to purchase dishes on the website, first, he needs to register an account by clicking "Register" button to enter the register page, and then just filling in User Name, Password and Repeat Password. If the customer register is successful, a page will display to inform the customer, also if there is a register fault, such as "User Name already exists", a friendly warning message will be given on the register page. Note that in the register form, there is a validation function to validate whether the information filled by the customer is valid or not. Only if all information is valid, the operation of register will be successful. a. Identify the functional and non-functional requirements for the above case study. (4)	K3	CO2	3.1.2 1.4.1 2.1.2 2.3.2																											

	b. Explain the elaboration task of requirement engineering process and visually represent it.(8)					
(Or)						
14.	a. Using the case study given in question 13, draw the data flow diagram. (4) b. Find the states and transition given in the case study, draw the state diagram and prepare the process activation table based on the various events. (8)	K3	CO2	3.1.2 1.4.1 2.1.2 2.3.2		
15.	a. Differentiate between the low and high data abstraction using suitable examples (8) b. Make use of the following design model to demonstrate how the low level and high level data abstraction are exhibited. (4)	<pre> graph TD Object --> Component Component --> Button Component --> Checkbox Component --> TextComponent TextComponent --> TextArea TextComponent --> TextField </pre>		K3	CO3	3.4.1 13.3.2 14.1.2
(Or)						
16.	a. Select the preferred architecture style for digital banking application and draw the architecture diagram. (4) b. Explain the five architectural styles used to design the software with suitable examples. (8)	K3	CO3	3.4.1 13.3.2 14.1.2		
17.	a. Explain any one technique of white box testing. (4) b. Differentiate between white and black box testing and elaborate on any one technique of black box testing. (8)	K2	CO4	2.2.1 13.3.2 14.1.2		
(Or)						
18.	a. Explain the importance of test cases in testing a software.(4) b. Differentiate between alpha and beta testing and how the validation testing ensures that the customer is satisfied by the developed product? (8)	K2	CO4	2.2.1 13.3.2 14.1.2		
19.	a. Outline the significance of scrum team. (4) b. How scrum uses technologies in software development projects to develop a product. (8)	K2	CO5	1.3.1 12.1.2		
(Or)						
20.	a. Summarize the steps involved in development to deployment of software. (4) b. Explain how finding the bottleneck activities can improve the production in DevOps. (8)	K2	CO5	1.3.1 12.1.2		